8.8.4 Vector Product

To define vector product between two vectors, we need the concept of right handed and left handed system.

If we align the fingers of our right hand along the vector $\vec{a}$ and bend our fingers around in the direction of rotation from $\vec{a}$ towards $\vec{b}$ (through an angle less than 180°), our thumb will point in the direction of $\vec{a}\times\vec{b}$. Now, following the right-hand rule, $\vec{b}\times\vec{a}$ will point in the direction opposite to $\vec{a}\times\vec{b}$ (See Fig. 8.38).

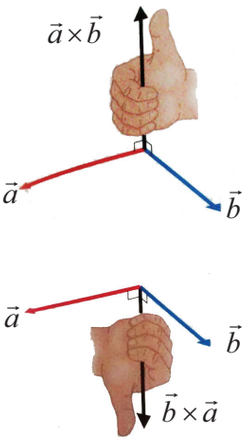
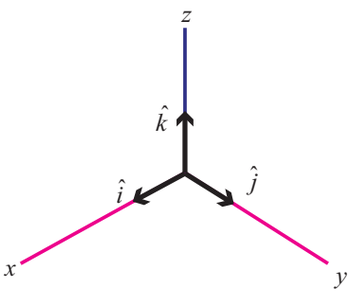


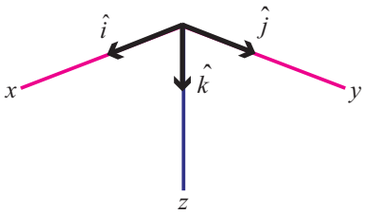
Fig. 8.38

We may also say that if $\vec{a}$ is rotated into the direction of $\vec{b}$ through the angle $\theta(<\pi)$, then $\vec{a}\times\vec{b}$ advances in the same direction as a right-handed screw would if turned in the same way.

A Cartesian coordinate system is called right-handed if the corresponding unit vectors $\hat{i},\hat{j},\hat{k}$ in the positive direction of the axes form a right-handed triple as in Fig. 8.39. The system is called left handed if the sense of $\hat{k}$ is reversed as in Fig 8.40.



Right handed
Fig. 8.39



Left handed
Fig. 8.40


Right handed screw
Fi. 8.41


Left handed screw
Fig. 8.42

Definition 8.17

Vector product of any two non-zero vectors $\vec{a}$ and $\vec{b}$ is written as $\vec{a}\times\vec{b}$ and is defined as

$$\vec{a}\times\vec{b} = |\vec{a}||\vec{b}|\sin\theta\,\hat{n},$$

where θ is the angle between $\vec{a}$ and $\vec{b}, 0\le\theta\le\pi$.

Here $\vec{a},\vec{b},\hat{n}$ form a right handed system.

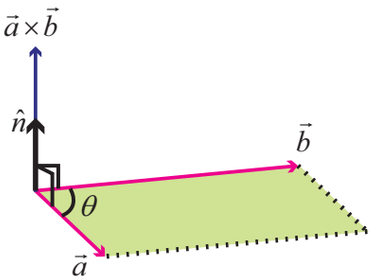


Fig. 8.43

The resultant is a vector with magnitude $|\vec{a}||\vec{b}|\sin\theta$ and has the direction $\hat{n}$.

Further $\vec{a}\times\vec{b}$ is a vector perpendicular to both $\vec{a}$ and $\vec{b}$. That is, $\vec{a}\times\vec{b}$ is normal to the plane containing $\vec{a}$ and $\vec{b}$.

Note 8.6

(i) Note that the order of the vectors is very important to decide the direction of $\hat{n}$.

(ii) Since the resultant is a vector, this product is named as vector product. Again, we use the symbol cross '×' to define such a product and hence it has another name **cross product**.

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